

# Using Deep Learning for Medical Automation in Diagnosis of Degenerative Arthritis

Sangmin Lee<sup>1</sup>, Namgi Kim<sup>2</sup>, Dohun Kim<sup>3</sup>

<sup>\*1</sup>Researcher, Dept. Computer Science, Kyonggi University, Republic of Korea

<sup>2,3</sup>Professor, Dept. Computer Science, Kyonggi University, Republic of Korea

d9249@kyonggi.ac.kr<sup>1</sup>, ngkim@kyonggi.ac.kr<sup>2</sup>, karmy01@kyonggi.ac.kr<sup>3</sup>

Corresponding author\*: Sangmin Lee

**Abstract** This study aimed to improve the accuracy of the classification of degenerative arthritis for the provision of a scientific basis and for automation through an artificial intelligence model in the degenerative arthritis field. The accuracy of classification was compared among deep learning models with excellent classification performance. Methods with randomness such as data augmentation were not applied to understand the objective performance of deep learning models. Specialized models of each class were identified through experiments, and it was confirmed that their accuracy was comparable to that of previous studies. Therefore, this study demonstrated the direction of future research and the possibility of automatic medical diagnosis through deep learning models.

**Keywords:** Knee osteoarthritis, deep learning, convolutional neural network, Kellgren-Lawrence grade, classification, knee X-ray

## 1. Introduction

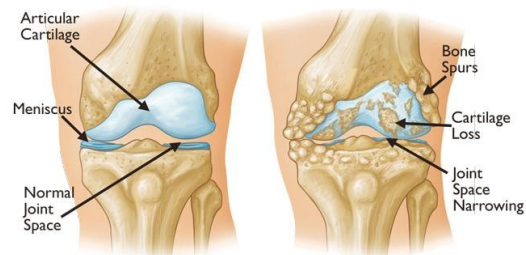
The human body is composed of many joints and bones. These joints, located between bones, play a role in facilitating the natural movement of the body. These are called cartilages, which play a crucial role in the movement of the body because the body cannot move freely if there is no cartilage or they have a problem. [1]

The symptoms of cartilage rupture may appear in anyone regardless of age. As shown in the example in Figure 1, symptoms of cartilage wear with age can be accompanied by local degenerative changes of articular cartilage, joint deformation, continuous pain, limited range of motion, and stiffness. These symptoms are generally referred to as knee osteoarthritis (OA). Basically there is a risk of infection and reoperation due to infection. Hence, aiming for improvement through non-surgical treatment rather than through surgical treatment is less burdensome on the body and mind. An accurate diagnosis of arthritis is crucial for early treatment to prevent surgical treatment.

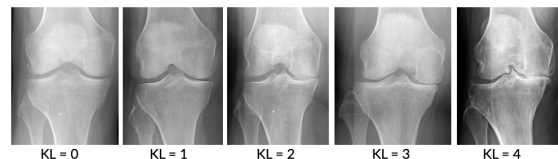
Simple X-ray imaging is performed in the diagnosis process of knee OA to base the visual changes as shown on the right side of Fig. 2.

These results are used to diagnose osteoarthritis, evaluate its severity, and determine its progress. However, the imaging process for diagnosis through simple X-rays and receiving the doctor's opinion takes much time and money for the patient. [2]. [3]

In this study, a deep learning model is used to diagnose knee OA to overcome the aforementioned problems. If high accuracy and reasoning results can be delivered to the doctor, the cost and time can be drastically reduced with an automated system based on a deep learning model by improving the apprenticeship process.



**Figure 1.** Example of knee artificial joint replacement [1]



**Figure 2.** Stage of knee osteoarthritis [2]

## 2. Related Works

As can be seen from Section I above, there is a growing need to improve diagnostic accuracy and methods of using deep learning for this purpose in the medical field are being actively researched. Previous studies compared image classification performance mainly by using

ResNet, VGG, and DenseNet. However, this study compared performance among Xception, MobileNet, and Transformer-based models, focusing on Xception and MobileNet, which had a significant impact among lightweight deep learning models, in order to understand the performance of more models and to understand the applicability of lightweight models. Furthermore, an accuracy comparison experiment was conducted to determine the accuracy of degenerative osteoarthritis problems as the Vision in transformer technique has been in the spotlight recently in the computer vision research field.

### 3. Dataset

The same knee X-ray data [4], which is generally used for comparison with previous studies, was used to understand the performance of each model through knee OA. The data distribution information is shown in Table 2.

**Table 2.** Composition Of Data Used In Training

|       | <i>Train</i> | <i>Test</i> | <i>Total</i> |
|-------|--------------|-------------|--------------|
| 0     | 2614         | 639         | 3253         |
| 1     | 1199         | 296         | 1495         |
| 2     | 1728         | 447         | 2175         |
| 3     | 863          | 223         | 1086         |
| 4     | 200          | 51          | 251          |
| Total | 6604         | 1656        | 8260         |

### 4. Experimental

As shown in Section 3, the K-fold training method was used because the total number of data was small. The entire data was equally divided into five sets. Four data sets were used for training and one data set was used for validation. The prediction was performed after each model was created. In the next step, the data that had been used for validation was used for training and the data that had not been used for validation was used for validation. This process was repeated five times. As a result, a total of five learning models were generated with the data divided into five folds for each model. Then, the performance of each model was inferred by generating one model using the average ensemble for the predicted values of the model.

**Table 3.** Composition Of Data Used In Training

| <i>Class</i> | <i>Model</i>     | <i>correct</i> | <i>Accuracy</i> |
|--------------|------------------|----------------|-----------------|
| 0            | TinyNetA         | 582/639        | 0.9108          |
| 1            | VGG19            | 149/296        | 0.5034          |
| 2            | DenseNet169      | 315/447        | 0.7047          |
| 3            | Swin_tiny_patch4 | 185/223        | 0.8296          |

|   |                 |       |        |
|---|-----------------|-------|--------|
|   | window7 224     |       |        |
| 4 | MobileVitV2 150 | 45/51 | 0.8824 |

### 5. Conclusion

This study investigated the deep learning application method to overcome the conventional apprenticeship method in the knee OA classification problem. The usability of CNN models, which were not used before, and the possibility of higher accuracy were demonstrated through performance comparison with previous studies and training of various models. This study did not use the augmentation technique that is generally used to solve the lack of data problem. However, the performance of existing studies could be improved if augmentation techniques or new methodologies are applied. In the future, we plan to conduct research to achieve higher accuracy using data augmentation techniques and various ensemble techniques.

### Acknowledgments

This research was supported by Basic Science Research Program through the National Research Foundation of Korea(NRF) funded by the Ministry of Education (No. 2020R1A6A1A03040583) and This work was supported by the GRRC program of Gyeonggi province. [GRRC KGU 2020-B04, Image/Network-based Intellectual Information Manufacturing Service Research]

### References

- [1] J. R. H. Foran, American Academy of Orthopaedic Surgeons, "Total knee replacement," June 2020. [Online]. Available: <https://orthoinfo.aaos.org/en/treatment/total-knee-replacement/>
- [2] N. Bayramoglu, M. T. Nieminen and S. Saarakkala, "A lightweight CNN and joint shape-joint space (JS2) descriptor for radiological osteoarthritis detection," Medical Image Understanding and Analysis, vol. 1248, pp. 331-345, 2020.
- [3] K. Jung, "Diagnosis and treatment of arthritis," Journal of the Korean Neurological Association, vol. 35, no. 4, 2017. <https://doi.org/10.17340/jkna.2017.4.28>
- [4] K. He, X. Zhang, S. Ren and J. Sun, "Deep residual learning for image recognition," arXiv preprint arXiv: 1512.03385, Dec 2015. <https://arxiv.org/abs/1512.03385>